

KALOYAN IVANOV

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PROFILE

I am a curiosity-driven Game Programming student at Saxion University of Applied Sciences with a strong background in **C++/OpenGL** and **C#/Unity**. I have experience in **Gameplay**, **Physics**, **AI**, and **Graphics programming**, with a solid foundation in **3D Mathematics**, **Software Architecture**, and **Design Patterns**. Bringing strong programming knowledge, I'm looking to gain hands-on experience in the creative industry and continue evolving as a developer.

COURSEWORK & SKILLS

Third Year

Sep 2025 - Jul 2026

- Pursued a self-directed Minor in **C++ Engine Development**, specializing in **Graphics Programming** with **OpenGL**, covering Rendering Pipelines, Shaders, and Engine Architecture.
- Completed a five-month **Internship** at Casualino JSC, where I was the **Primary Gameplay Programmer** responsible for **implementing Core Gameplay Systems** for a **Mobile Puzzle Game** in Unity Engine (C#), while collaborating within a multidisciplinary development team.

Second Year

Sep 2024 - Jul 2025

- Pursued advanced coursework in Graphics programming, with a focus on **3D Mathematics**, **C++**, and **3D Rendering using OpenGL**.
- Familiarized myself with more **Software Architecture** techniques, by applying **SOLID principles** and various **design patterns** while developing a **Unity 3D Tower Defense game**.

First Year

Sep 2023 - Jul 2024

- Learned how to apply my **C# knowledge** to **gameplay**, **physics** and **algorithms programming** by utilizing a custom-made 2D game engine by the teachers.
- Developed my first fully functional **Unity 3D Game**, applying various **Software Architecture** techniques to ensure **maintainability** and **scalability**.
- Completed supplementary courses covering **Game** and **UI/UX design**, as well as professional skills like **Communication** and **Research**.

Group Projects

- Collaborated on multiple **group projects** (2-3 weeks), working closely with **Engineers**, **Designers**, and **Artists** to develop unique games.
 - **Effective communication** and **strong teamwork** allowed us to **navigate challenges** in a fast-paced development cycle and **successfully achieve our goals**.
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TECHNICAL SKILLS

- **Programming Languages:** C++, OpenGL (GLSL), C#
- **Tools:** Unity, SFML, GXPEngine (developed by teachers), Git, Jira, Visual Studio, Rider
- **Additional Technologies:** Figma, Tiled, Java, HTML, CSS (SASS), JavaScript (jQuery), MS SQL, ASP.NET Core MVC, Elixir, Erlang

PROJECTS & EXPERIENCE

Mobile Puzzle Game - Internship Project at Casualino JSC, Varna

Feb - Jun 2026

- Collaborated within a **multidisciplinary team** of Game Designers, UI/UX Designers, QA, and Project Managers.
- Conducted **Competitor Research** during the project's Research Phase alongside a fellow intern to identify **Gameplay Mechanics** and **Design Opportunities**.
- Primarily responsible for implementing **Core Gameplay** and **Player Progression Systems** during the Prototyping and Development Phases, while following Clean Coding Practices.

ZRenIE - A 3D C++ OpenGL Engine

Sep 2025 - Present

- **ZRenIE** is a personal project I started for learning purposes, with a focus on **exploring engine development** and **deepening my C++ knowledge**.
- Its purpose is to serve as a flexible tool for experimenting with modern rendering techniques - **abstracting** away irrelevant details while allowing for **easy modification** and **extension** of the rendering pipeline.
- To achieve this, I am **designing** the architecture around **modularity**, making it **straightforward** to integrate new systems or behaviors without disrupting existing components.

3D Rendering with OpenGL

Nov 2024 - Present

- Independently studying OpenGL to expand my knowledge of real-time rendering.
- Thoroughly worked through the first two chapters of the 'LearnOpenGL' book, focusing on **Coordinate Spaces**, **Camera Behavior**, and the **Phong Lighting Model**.
- Developed an interactive scene with Unity-style navigation, featuring real-time lighting from **Directional**, **Point** and **Spot** lights.

Procedural Paris - Procedural recreation of Paris in Unity

Oct 2025 - Present

- A project in which I experiment with different **Procedural Generation techniques** to **recreate** a small-scale city clone of the scenic **French Revolution Paris** of Assassin's Creed Unity.
- Procedural Generation techniques include **modular meshes** and **shape grammar**, **editor tooling**, and **mesh generation**.
- Conducted **Visual Research** on Paris in 1789 and documented key insights.

C++

Sep - Nov 2024

- Participated in a **C++** course, exploring key concepts such as **memory management** through pointers and references, **CONST-ness**, OOP principles like **constructors** and **destructors**, **File I/O**.
- Designed and developed a 2D Battle Game using the **SFML framework**, implementing game object hierarchy, scene setup by mapping JSON data into game objects, **sprite animations**, and persistent **high-score tracking** between sessions.
- Scored **10/10** on my final assessment.

HellFryer - Unity 3D Split-Screen Co-op Game

Jan 2025

- Developed as part of a 3-week group project with a team of 6, including Engineers, Designers, and Artists.
- Established a clear **Git workflow** and code architecture, ensuring maintainability using **SOLID principles** and **design patterns**.
- Responsible for implementing the **co-op gameplay**, **role-switching mechanics**, **hunting equipment**, **monster behaviors**, and part of the **hostile items' interactions**.
- Successfully managed project complexity, meeting all planned milestones within the deadline.
- Achieved a final project score of **9/10** due to strong **teamwork**, **organization**, and **execution**.

EDUCATION

Bachelor of Creative Media and Game Technologies

Sep 2023 - Jul 2027 (Expected)

Saxion University of Applied Sciences

- Specialization: Game Engineer - Focus on Game programming

Mathematics and Computer Science

Sep 2018 - Jun 2023

High School of Mathematics "Dr. Petar Beron"

ADDITIONAL EXPERIENCE

Functional programming Software Development

Aug - Dec 2022

Internship at Quanterall Academy

- Erlang and Elixir software development

C# Web development

Feb 2021 - Nov 2022

Software University (SoftUni)

- C# with MS SQL and ASP.NET Core MVC framework software development.
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LANGUAGES

English - C1

Bulgarian - Native

IMAGES

